Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Business Innovation with Internet-of-Things (IoT)

TGS Ref No: TGS-2026064711

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 10.1

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
9.0	1 January 2026	Previous release — 2-day structure with ThingSpeak/MATLAB and IFTTT practical activities.	Dr. Alfred Ang
10.0	8 August 2026	Major revision. Restructured to the accredited one-day (8-hour) format per Course Proposal CA-WSQ-2020-007388. All practical activities replaced with four hands-on labs on the IoTFlow platform (https://iot.tertiaryinfotech.com); ThingSpeak, MATLAB and IFTTT content retired. Assessment stated as WA (Q&A) plus Oral Questioning.	Dr. Alfred Ang
10.1	11 August 2026	Each lab is now a self-contained folder (labs/lab-NN/) holding its instructions plus runnable Python scripts built on the official iotflow client. No change to the schedule, timings or assessment.	Dr. Alfred Ang

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Course Information

Course Title	Business Innovation with Internet-of-Things (IoT)
WSQ Course Reference	TGS-2026064711
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
TSC Title / Code	Business Innovation · ICT-SNA-4003-1.1 (Level 4)
Duration	1 day · 8 training hours (excluding a 1-hour lunch break)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks counted within training time)
Course Components	Classroom Facilitation 5.0 h · Practical / Practicum 1.5 h · Assessment 1.5 h
Mode	Instructor-led classroom facilitation with hands-on IoT labs on the IoTFlow cloud platform
Platform	IoTFlow — https://iot.tertiaryinfotech.com
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Compare the organisation's current business model with emerging IoT-enabled business models in the industry. (A1, K1)
- LO2: Explore potential opportunities for business innovation using IoT across industry use cases. (A2, K1)
- LO3: Identify how IoT works and the ways digitisation can be applied to the business. (A3, K2)
- LO4: Conduct a feasibility analysis and weigh the costs and benefits of adopting IoT. (A4, K3)
- LO5: Implement business innovation processes with IoT in line with a defined action plan. (A5, K4)

Skills Framework Mapping (TSC ICT-SNA-4003-1.1 Business Innovation)

Identify and evaluate digitisation and innovative business opportunities provided by new advancements in information and communication technology to establish new services or businesses to bridge the physical and digital worlds.

Abilities (assessed by the Written Assessment)	Knowledge (assessed by Oral Questioning)
A1 — Compare current business model for the organisation with the other business models in the industry	**K1 —** New and emerging innovative business processes / models deployed in the industry
A2 — Explore potential opportunities for	**K2 —** Concept and principles of digitisation

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business innovation to be introduced within the organisation	
A3 — Identify ways in which digitisation can be applied to the business	**K3 —** Factors to consider to assess the viability of business innovation for a given organisation context
A4 — Conduct feasibility analysis and weigh the costs-benefits of potential business innovation opportunities	**K4 —** Success factors for implementation of business innovation
A5 — Implement business innovation processes in line with a defined action plan	

Assessment

- Written Assessment (WA) — Short-Answer Questions (Q&A), 75 minutes, open book. Covers A1–A5.
- Oral Questioning (OQ) — 15 minutes, 1:1 with the assessor, open book. Covers K1–K4.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The assessment is conducted at the end of the training day, from 5:00 pm.
- Assessor-to-learner ratio: 1:3 to 1:10 for the WA (Q&A); 1:1 for Oral Questioning.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — IoT Business Innovation — Opportunity, Technology, Adoption & Implementation

Time	Duration	Topic / Activity
9:30am–9:45am	15 min	Welcome, trainer and learner introductions, ground rules, and mandatory digital attendance (AM)
9:45am–9:55am	10 min	Learning outcomes, course outline and Skills Framework (TSC ICT-SNA-4003-1.1) briefing
9:55am–10:50am	55 min	**Topic 1 — Overview of Internet of Things (IoT): Industry 4.0, IoT building blocks, sensors and actuators, wireless technologies, and the new business models IoT enables. Includes the facilitated discussion 'Compare Business Models'. (LO1 / A1, K1)**
10:50am–11:05am	15 min	Tea break
11:05am–12:00pm	55 min	**Topic 2 — Potential Opportunities of IoT Innovation: use cases across agriculture, autonomous vehicles, logistics, smart**

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		buildings, healthcare and drones, and a repeatable method for finding an opportunity. Includes the facilitated discussion 'Spot an IoT Opportunity'. (LO2 / A2, K1)
12:00pm–12:30pm	30 min	**Topic 3 — IoT Technology (concepts, part 1): cloud computing for IoT, the IoTFlow platform, the IoT data round trip, and uplink versus downlink. (LO3 / A3, K2)**
12:30pm–1:30pm	60 min	Lunch break
1:30pm–1:40pm	10 min	Mandatory digital attendance (PM)
1:40pm–1:55pm	15 min	**Topic 3 — IoT Technology (concepts, part 2): turning data into business insight — visualise, detect, notify, record, analyse with AI and act automatically.**
1:55pm–3:25pm	90 min	**Topic 3 — HANDS-ON PRACTICAL on IoTFlow (https://iot.tertiaryinfotech.com). Lab 1: Register Your First IoT Device on IoTFlow; Lab 2: Post Business Data to the Cloud (HTTP & MQTT); Lab 3: Read Data Back and Remotely Control the Asset; Lab 4: Turn IoT Data into Business Insight**
3:25pm–3:40pm	15 min	Tea break
3:40pm–4:20pm	40 min	**Topic 4 — IoT Adoption: adoption barriers, why IoT security differs, the Mirai botnet case study, edge versus cloud processing, and cost-benefit analysis. Includes the facilitated discussion 'Cost-Benefit Analysis'. (LO4 / A4, K3)**
4:20pm–4:45pm	25 min	**Topic 5 — IoT Implementation: the six success factors, the Stanford d.school design-thinking model, and the 7-step implementation action plan. Includes the facilitated discussion 'Your Implementation Action Plan'. (LO5 / A5, K4)**
4:45pm–5:00pm	15 min	Course recap and Q&A, TRAQOM survey, Assessment digital attendance, and Briefing for Assessment

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5:00pm–6:15pm	75 min	**Written Assessment (WA) — Short-Answer Questions (Q&A), 75 minutes, open book. Assesses A1–A5 (15 minutes per ability).**
6:15pm–6:30pm	15 min	**Oral Questioning (OQ) — 15 minutes, 1:1 with the assessor, open book. Assesses K1–K4.**

Total training time: 480 minutes (8 hours), excluding the 1-hour lunch break. Classroom Facilitation 300 min (5.0 h) · Practical/Practicum 90 min (1.5 h) · Assessment 90 min (1.5 h).

Topic and Activity Reference

Topic / Learning Unit	Outcome & TSC mapping	In-class activity
Topic 01: Overview of Internet of Things (IoT)	LO1 · A1, K1 · 60 min	Discussion — Compare Business Models (15 min facilitated discussion)
Topic 02: Potential Opportunities of IoT Innovation	LO2 · A2, K1 · 60 min	Discussion — Spot an IoT Opportunity (15 min facilitated discussion)
Topic 03: IoT Technology	LO3 · A3, K2 · 60 min + 90 min practical	Lab 1: Register Your First IoT Device on IoTFlow, Lab 2: Post Business Data to the Cloud (HTTP & MQTT), Lab 3: Read Data Back and Remotely Control the Asset, Lab 4: Turn IoT Data into Business Insight
Topic 04: IoT Adoption	LO4 · A4, K3 · 60 min	Discussion — Cost-Benefit Analysis (20 min facilitated discussion)
Topic 05: IoT Implementation	LO5 · A5, K4 · 60 min	Discussion — Your Implementation Action Plan (20 min facilitated discussion)

Hands-On Labs (Topic 3 — Practical / Practicum, 90 minutes)

All labs run on IoTFlow (https://iot.tertiaryinfotech.com). Full step-by-step instructions are in the Learner Guide; the slide deck presents each lab's purpose and outcome only.

Lab	Title	What the learner builds
Lab 1	Register Your First IoT Device on IoTFlow	A registered IoT device in your own IoTFlow project, with its device token saved
Lab 2	Post Business Data to the Cloud (HTTP & MQTT)	Live temperature and humidity telemetry streaming into IoTFlow from terminal and Python
Lab 3	Read Data Back and Remotely Control the Asset	A dashboard switch and slider that control a relay/LED on the device in real time
Lab 4	Turn IoT Data into Business Insight	A monitoring dashboard plus an alert-driven n8n workflow with an AI summary and a downlink command

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Instructional Methods

- Lecture — trainer-led explanation of concepts with visual slides.
- Didactic questioning — the trainer poses questions to check and deepen understanding.
- Facilitated discussion — learners apply each topic to their own organisation (Topics 1, 2, 4 and 5).
- Demonstration — the trainer demonstrates each IoTFlow workflow before the learners attempt it.
- Hands-on practical — learners complete Labs 1–4 on their own IoTFlow project (Topic 3).

Resources Required

- Classroom with projector or large screen, whiteboard and high-speed WiFi.
- One laptop per learner with a modern web browser (spare laptops are available).
- A free learner account on IoTFlow (https://iot.tertiaryinfotech.com).
- Python 3 with the IoTFlow client (pip install "iotflow[mqtt]") for the optional scripted labs.
- IoT hardware (ESP32/ESP8266) is optional — every lab can be completed from the browser and terminal.
- Course slides, Learner Guide and Lesson Plan available on https://lms-tms.tertiaryinfotech.com.

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